- Let
  - W = Mean waiting time
  - N = Mean number of customers in the queue
  - 1/ μ = Mean service time
  - R = Mean residual service time
- We can prove that
  - W = N * (1/ μ) + R
  - W = N E[S] + R

- Applying Little's Law to the queue
  - N = λ W

Mean residual time R

$$R = \frac{1}{2}\lambda E[S^2]$$

Waiting time
= #Customers in front * Mean Service time + Residual Service time

$$W = \frac{R}{1-\rho}$$

$$\rho = \lambda E[S]$$

Priority	E[S]	E[S^2]	Arrival rate	$\rho = \lambda$ E{S]	Residual service time
1	0.5	0.375			
2	0.4	0.400			
3	0.3	0.180			

$$W_1 = \frac{R}{1-\rho_1}$$

$$W_2 = ?$$

$$W_3 = ?$$

$$\rho_1 = \lambda_1 E[S_1]$$
$$\rho_2 = \lambda_2 E[S_2]$$

$$\rho_3 = \lambda_3 E[S_3]$$

$$R = ?$$

Priority	E[S]	E[S^2]	Arrival rate λ_i	ρ = λ E{S]	Residual service time
1	0.5	0.375	1.2*0.5=0.6		
2	0.4	0.400	1.2*0.3=0.36		
3	0.3	0.180	1.2*0.2=0.24		

$$W_1 = \frac{R}{1-\rho_1}$$

$$W_2 = \frac{R}{(1-\rho_1)(1-\rho_1-\rho_2)}$$

$$W_3 = \frac{R}{(1-\rho_1-\rho_2)(1-\rho_1-\rho_2-\rho_3)}$$

$$\rho_1 = \lambda_1 E[S_1]$$
$$\rho_2 = \lambda_2 E[S_2] \qquad \rho_3 = \lambda_3 E[S_3]$$

$$R = \tfrac{1}{2}\lambda_1 E[S_1^2] + \tfrac{1}{2}\lambda_2 E[S_2^2] + \tfrac{1}{2}\lambda_3 E[S_3^2]$$